

user experience. Under ERC-1155 and similar to ERC-721, operators are approved for all supported tokens in a binary all-or-none fashion.

CUSTODY

A critical DeFi primitive is the ability to escrow or custody funds directly in a smart contract. This is distinct from the situation in ERC-20 when operators are approved to transfer a user's balance. In that case, the user still retains custody of their funds and could transfer the balance anytime or revoke the contract's approval. When a smart contract has full custody over funds, it presents the possibility for new capabilities (and additional primitives), including:

- Retaining fees and disbursing incentives
- Facilitation of token swaps
- Market making of a bonding curve
- Collateralized loans
- Auctions
- Insurance funds

To effectively custody tokens, a contract must be programmed to handle the interface of the corresponding type, which would be ERC-20 for fungible and ERC-721 for non-fungible. The contract could generically handle all tokens of that interface or of a specific subset only. When a token is sent to a contract, it could become permanently custodied if the contract has no encoded mechanism for releasing